Social Media Usage and Mental Health Risk Prediction

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I. Introduction

Social media is a significant part of the lives of the youth and students that promotes communication and sharing of information via social networking sites and applications such as Instagram, Facebook, Tik Tok, and Twitter. Nonetheless, frequent use of these platforms can have a negative effect on mental health, causing the development of such problems as loss of sleep, anxiety, and other mental issues. The recent studies highlight the importance of investigating the relation between social media use and the risk of mental illness. To examine such a relationship, data science and machine learning are used to process behavioral data, such as the number of hours per day on social media, the number of hours to sleep, and the amount of time spent with the screen to determine the ability to predict the risk of depression. The project is based on a small dataset that contains user demographics age, gender, sleep hours, social media hours, and in which the exploratory data analysis is performed, and a Random Forest classification model has been implemented with the purpose of predicting the categories of depression risks. The findings of the project indicate that the use of behavioral data to reveal the mental health trends and the fact that machine learning methods may help predict the possibility of mental health issues.

II. Problem Formulation

The project investigates the effect of the use of social media on mental health, especially its association with the risk of depression. It outlines that too much screen time and lack of sleep can increase stress and symptoms of depression. The proposed research will apply data analysis and machine learning to examine the usage patterns and determine the behavioral clues of mental health risks to assess a dataset which contains the user demographics and social media habits.

The key research question of this study is: *Is it possible to predict the degree of depression risk based on the patterns of social media use with machine learning tools?*

To apply a machine learning classification model predicting the risk of depression: Low, Medium, and High Risk, the exploratory analysis of data is performed to comprehend the pattern between variables. The paper looks at the extent to which behavioral information on the use of social media can be useful information in predicting a mental health risk.

III. Data Acquisition

In as far as this project is concerned, a structured dataset addressing social media usage and indicators of mental health was used. The data presented in a CSV format consists of 120 records and 6 attributes providing the information on the daily social media habits of the users and their basic demographics. The hypothesis of the analysis is to indicate whether the trends in the use of social media can be used to predict the level of depression risk. A dataset was manipulated with the help of the Pandas library it is a popular data science library to manipulate and analyze structured data. The variables included in the dataset are shown in Table I.

TABLE I: Dataset Variables

Feature	Description
Age	Age of the user
Gender	Gender of the user
Daily Usage Hours	Average number of hours spent on social media per day
Sleep Hours	Average number of sleep hours per day
Screen Time	Total screen time per day
Depression Risk	Target variable indicating mental health risk (Low, Medium, High)

Once the dataset was loaded, the head () function was used to show the first few rows to ensure that the dataset had been imported properly

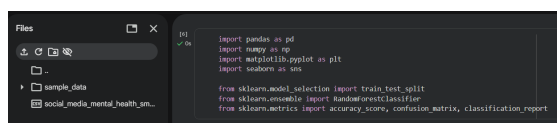
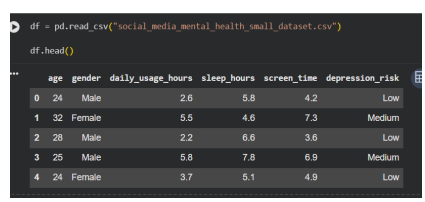


Fig. 1: Importing Python libraries required for data analysis and machine learning.



```
df = pd.read_csv("social_media_mental_health_small_dataset.csv")
df.head()
```

	age	gender	daily_usage_hours	sleep_hours	screen_time	depression_risk
0	24	Male	2.6	5.8	4.2	Low
1	32	Female	5.5	4.6	7.3	Medium
2	28	Male	2.2	6.6	3.6	Low
3	25	Male	5.8	7.8	6.9	Medium
4	24	Female	3.7	5.1	4.9	Low

Fig. 2: Loading the social media mental health dataset using Pandas.

The shape of the dataset was also examined to understand its size.

```
df.shape
(120, 6)
```

Fig. 3: Dataset shape showing 120 records and 6 attributes.

Also, the dataset structure was examined with the help of the info () function to ensure that there are no missing values and the data types of each column.

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 6 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   age                120 non-null    int64
 1   gender             120 non-null    object
 2   daily_usage_hours  120 non-null    float64
 3   sleep_hours        120 non-null    float64
 4   screen_time        120 non-null    float64
 5   depression_risk    120 non-null    object
dtypes: float64(3), int64(1), object(2)
memory usage: 5.8+ KB
```

Fig. 4: Dataset structure showing column data types and non-null values.

The describe () function was also used to give statistical summaries of the numerical variables. Such a step gave practical information, including the average age of users, average number of hours using social media, and average number of hours sleeping.

Fig. 5: Statistical summary of numerical attributes.

With such steps, the dataset was loaded and explored successfully and gave a clear understanding of the variables to be analyzed further.

IV. Data Cleaning and Preprocessing

Data cleaning is an essential step in data analysis, as well as machine learning to guarantee the absence of errors in the dataset and improves the credibility of finding. To test the missing values in Python, the `isnull()` method is used because otherwise it may adversely affect the model performance and cause false analysis.

```
df.isnull().sum()

                    0
age                 0
gender              0
daily_usage_hours   0
sleep_hours         0
screen_time         0
depression_risk     0

dtype: int64
```

Fig. 6: Checking for missing values in the dataset.

The outputs showed that the data set is complete and that no values are missing in any of the columns and so it can be analyzed further. Also, the dataset was tested on the number of repeats of the records with the help of `duplicated()` function, since repeats may cause prejudice in the findings

```
df.duplicated().sum()
np.int64(0)
```

Fig. 7: Checking for duplicate entries.

The result of the output ensured that no duplicates existed in the dataset. Nonetheless, the duplicate removal command was still implemented to make sure that the data is consistent.

```
df.drop_duplicates(inplace=True)
```

Fig. 8: Removing duplicate records from the dataset.

Because the dataset was not missing any value or had duplicates, the dataset was clean without carrying out any major cleaning procedure. This pre-processing enabled the dataset to be prepared to be processed into the next stages of the exploratory data analysis (EDA) and machine learning modeling.

V. Exploratory Data Analysis (EDA)

The data analysis process is significant, and EDA is one of the steps to be taken. It aids researchers in the description of how the data looks, finding patterns, and analyzing associations among variables. Within the framework of this project, several visualizations were designed to examine the patterns of mental health indicators in relation to the use of social media. The dataset was investigated using different graphs like histograms, scatter plots, count plots, and correlation heatmaps.

A. Daily Social Media Usage Distribution

A histogram was plotted to determine the duration that the users spend on social media by displaying the frequency of hours spent on social media per day.

```
plt.figure(figsize=(6,4))
sns.histplot(df['daily_usage_hours'], bins=10, kde=True)
plt.title("Daily Social Media Usage Distribution")
plt.xlabel("Hours per day")
plt.show()
```

Fig. 9: Python code used to generate the histogram.

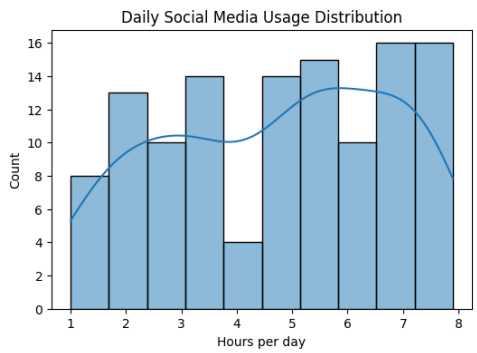


Fig. 10: Distribution of daily social media usage hours.

The histogram indicates that many people spend 3 to 7 hours daily on social media sites. It means that moderate and high social media consumption is prevalent among the users in this dataset. Few users dedicate very low or very high hours to social media.

B. Social Media Usage vs Sleep Hours

The correlation between the use of social media and the duration of sleep was plotted in a scatter plot.

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='daily_usage_hours', y='sleep_hours', data=df)
plt.title("Social Media Usage vs Sleep Hours")
plt.show()
```

Fig. 11: Code used to generate the scatter plot.

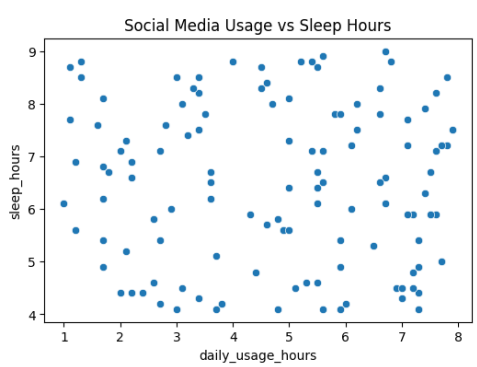


Fig. 12: Scatter plot showing the relationship between social media usage and sleep hours.

The scatter plot is used to depict the hours spent on social media and the hours spent asleep by the users per day. The graph indicates that people that spend more time on social media might occasionally have marginally reduced sleep hours. However, not a linear relationship which means that other variables can also affect sleep patterns.

C. Depression Risk Distribution

The number of users who belong to each category of depression risks were observed using a count plot.

```
plt.figure(figsize=(6,4))
sns.countplot(x='depression_risk', data=df)
plt.title("Depression Risk Distribution")
plt.show()
```

Fig. 13: Code used to generate the depression risk distribution chart.

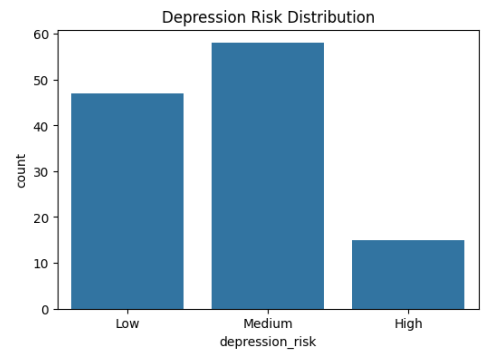


Fig. 14: Distribution of depression risk levels.

The count plot indicates the number of users in the Low Risk, Medium Risk and High-Risk categories. The findings show that the Medium Risk category has the maximum number of users than the Low-Risk category. The category of High Risk has the least number of individuals. This distribution indicates that many of the users in the dataset are in the average-risk group of mental health.

D. Correlation Analysis

To test the correlation amongst numerical variables, a heatmap of correlation was created.

```
plt.figure(figsize=(6,4))
sns.heatmap(df[['daily_usage_hours','sleep_hours','screen_time','age']].corr(), annot=True)
plt.title("Correlation Matrix")
plt.show()
```

Fig. 15: Code used to generate the correlation heatmap.

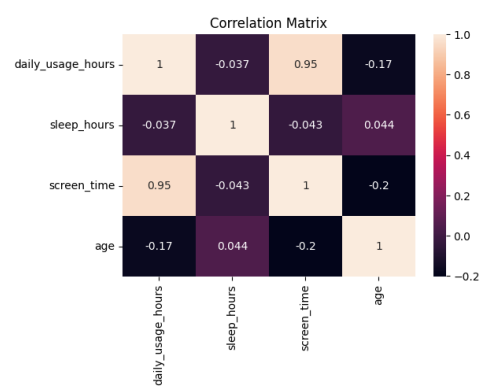


Fig. 16: Correlation matrix showing relationships between variables.

The heatmap indicates that there is a strong positive correlation between daily usage hours and screen time, which indicates that the more time spent on social media, the more the screen time. Conversely, age and sleep hours exhibit less significant correlated relationships with the use of social media. The exploratory data analysis provided the necessary information regarding the user behavior and the factors which could influence the level of mental health risks.

VI. Machine Learning Model

Social media usage patterns were used in developing a machine learning model that was based on the Random Forest Classifier to predict the level of depression risk in this project. It is an supervised machine learning algorithm that creates numerous decision trees to boost classification accuracy and apply to complicated data sets and reduce overfitting. The model is designed to classify the depression risk of users in terms of numerous behavioral and demographic characteristics.

A. Feature Selection

The previously outlined features were Daily Usage Hours, Sleep Hours, Screen Time, and Age, whereas the target variable to be predicted was Depression Risk, and the model was trained before.

```
X = df[['daily_usage_hours','sleep_hours','screen_time','age']]
y = df['depression_risk']
```

Fig. 17: Selecting input features and target variable for the machine learning model.

These characteristics were selected as they depict behavioral patterns that can put one in danger of mental health.

B. Train-Test Split

To evaluate the performance of the model, it was divided into two parts, 80 percent to be trained and 20 percent to be tested. The machine learning model was trained using the training dataset and its performance on unknown data assessed using the testing dataset.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Fig. 18: Splitting the dataset into training and testing sets.

This approach helps prevent overfitting and ensures that the model can generalize well to new data.

C. Model Training

The classifier developed by the Random Forest algorithm that is trained on the data set learns trends in terms of input features and constructs a decision tree to classify the levels of risk of depression. The model focuses on the relationships between social media use, sleep hours and screen time to determine their influence on the risk of depression.

```
model = RandomForestClassifier()

model.fit(X_train, y_train)

RandomForestClassifier
RandomForestClassifier()
```

Fig. 19: Training the Random Forest classifier.

D. Model Prediction

The predictions were then made after the model has been trained with testing data. The model is based on the acquired patterns to assign each test sample to one of the categories of risk of depression. The actual values are then compared with these predicted values, and this helps to assess the performance of the model.

```
predictions = model.predict(X_test)
```

Fig. 20: Generating predictions using the trained model.

VII. Model Evaluation

Once the Random Forest classifier has been trained, evaluation of the model is conducted to determine the effectiveness of the model to predict the categories of depression risks. This is an important step in machine learning to measure the accuracy and reliability of predictions, where one uses accuracy score, the confusion matrix and classification report.

A. Model Accuracy

The score of accuracy of about 0.958 (95.8 percent), has been used to show that the machine learning model was able to classify most samples in the test data, meaning that the chosen features are useful in predicting the level of risk of depression.

```
accuracy = accuracy_score(y_test, predictions)

print("Model Accuracy:", accuracy)

Model Accuracy: 0.9583333333333334
```

Fig. 21: Model accuracy obtained from the Random Forest classifier.

B. Confusion Matrix

The confusion matrix was used to determine the performance of the classification model, which has shown that most of the predictions are correct, particularly in the Low and Medium Risk categories with fewer errors in the High-Risk category. This implies that the Random Forest classifier is applicable in this dataset.

```
cm = confusion_matrix(y_test, predictions)

sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.show()
```

Fig. 22: Code used to generate the confusion matrix.

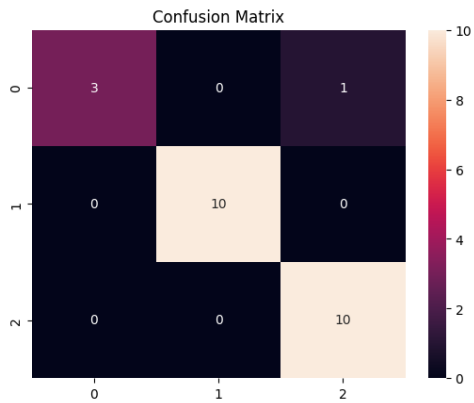


Fig. 23: Confusion matrix showing classification performance.

C. Classification Report

An analysis report indicated that the model has a high performance on all depression risk categories. Among the major findings is a perfect precision and recall of Low-Risk category, high recall of Medium Risk category, and a satisfactory performance of the High-Risk category, though slightly lower than the recall. The weighted average F1-score is nearly 0.96, which means that there is successful classification of the levels of depression risk.

```
print(classification_report(y_test, predictions))

              precision    recall  f1-score   support

        High       1.00      0.75      0.86         4
         Low       1.00      1.00      1.00        10
      Medium       0.91      1.00      0.95        10

    accuracy                           0.96        24
   macro avg       0.97      0.92      0.94        24
weighted avg       0.96      0.96      0.96        24
```

Fig. 24: Classification report displaying precision, recall, and F1-score.

VIII. Results Interpretation

The machine learning model evaluated the relationship between the use of social media and the level of depression risk using such variables as hours individuals spend on social media per day per day, the duration of sleep, hours spent on the screen, and age. It got a high accuracy of about 95.8 percent which depicts the potential of the model to categorize the level of mental health risks correctly. The main results comprised a significant connection between the use of social media and the length of the screen time, and the connection between the high usage and the shortness of the sleep period, which could reflect greater mental health risks. The model worked exceptionally well in detecting users under the Low and Medium Risk groups but there was a minor decrease in recall of users who belonged to the High-Risk group. In general, these findings point to the future potential of behavioral predictors such as social media use and sleep patterns to predict mental health risks. Although the dataset has its flaws and limitations, the results indicate that machine learning may be useful in identifying mental health concerns and that it may be involved in creating early detection systems or online wellness trackers in the future.

IX. Reflection

This project was aimed at using data analysis and machine learning to test the connection between social media and mental health. The most important processes of the data science workflow, such as data acquisition, preprocessing, exploratory data analysis (EDA), model development, and evaluation, have been discussed. Through EDA, patterns and relationships between variables including social media usage, sleep duration and screen time were identified through visualization, including histograms and scatter plots. Random Forest classifier was chosen to predict the risk of depression because it is effective when it comes to dealing with more than two features. The project also increased the hands-on experience of using Python libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn. In general, it highlighted the role of data-driven methodologies in dealing with social and health-related problems.

X. Conclusion

This project discussed the relationship between the use of social media and the risk of depression using data analysis and machine learning. A user demographics and social media behavior dataset were cleansed and examined using a range of visual tools and found a significant correlation between screen time and social media usage, with a certain deviation in the number of sleep hours. A random forest model has been constructed with an approximation of 95.8 percent accuracy in the prediction of categories of depression risks. The results indicate that social media behavior may provide useful information on the risk of mental health, but further research should involve larger samples and other behavioral indicators to provide greater accuracy.